

VOLTAMMETRY AND POLAROGRAPHY

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14.1. INTRODUCTION

Voltammetry is a group of electroanalytical techniques which deals with the interplay between electricity and chemistry, namely, the measurements of electrical quantities, such as current, potential, or charge and their relationship to chemical parameters. Such use of electrical measurements for analytical purposes has found a vast range of applications, including environmental monitoring, industrial quality control, or biomedical analysis. In contrast to many chemical measurements, which involve homogeneous bulk solutions, electrochemical processes take place at the electrode-solution interface. The term voltammetry is derived from electrical parameters/units, volt-am (pere)-metry. Voltammetry may be defined as "*the study of current voltage relation at a micro electrode, called working electrode*". In other words, the technique is based upon the potential current behavior of a polarizable electrode in the solution being analyzed. According to the recommendation of IUPAC, voltammetry is the general term to be used when the current potential relationships are being investigated. The surface area of voltammetry electrode is very small (0.0001 m^2) which is helpful in providing response of current produced by species in solutions. It is also important to clarify here that when mercury (flowing through a fine capillary seems to be emerging as a droplet) is used as a small electrode in voltammetric cell, the technique has a specific name, *polarography*.

14.2. PRINCIPLE OF VOLTAMMETRY

All the electrochemical techniques involve the use of electricity either as an input or output signal. The principle of voltammetry technique is to generate an input electrical signal and to measure corresponding output electrical signal. The input signals can be a preprogrammed function of time whereas the output signals may change with time due to sample under investigation. The signals are typically voltage, current and charge where charge is the integration of current with time. The input electrical signals usually cause the electrochemical reactions at working and counter electrodes. Simply, a measured small potential is established across a pair of electrodes, one of which is a non-polarizable reference electrode and the other a polarizable inert electrode. The current which flows depends upon composition of the solution.

Voltammogram

A plot of current as a function of applied potential is called a Voltammogram. It is the electrochemical equivalent of a spectrum in spectroscopy, providing quantitative and qualitative information about the species involved in the oxidation or reduction reaction



where O and R are the oxidized and reduced forms, respectively, of the redox couple. Such a reaction will occur in a potential region that makes the electron transfer

thermodynamically or kinetically favorable. For systems controlled by the laws of thermodynamics, the potential of the electrode can be used to establish the concentration of the electroactive species at the surface $[C_O(0,t)]$ and $[C_R(0,t)]$ according to the Nernst equation. On the negative side of E° , the oxidized form thus tends to be reduced, and the forward reaction (i.e., reduction) is more favorable. The current resulting from a change in oxidation state of the electroactive species is termed the *faradaic current* because it obeys Faraday's law (i.e., the reaction of 1 mol of substance involves a change of $n \times 96,487\text{C}$). The faradaic current is a direct measure of the rate of the redox reaction.

The pathway of the electrode reaction can be quite complicated, and takes place in a sequence that involves several steps. The rate of such reactions is determined by the slowest step in the sequence. Simple reactions involve only mass transport of the electroactive species to the electrode surface, electron transfer across the interface, and transport of the product back to the bulk solution. More complex reactions include additional chemical and surface processes that either precede or follow the actual electron transfer. The net rate of the reaction, and hence the measured current, may be limited by either mass transport of the reactant or the rate of electron transfer. The more sluggish process will be the rate-determining step. Whether a given reaction is controlled by mass transport or electron transfer is usually determined by the type of compound being measured and by various experimental conditions (electrode material, media, operating potential, mode of mass transport, time scale, etc.).

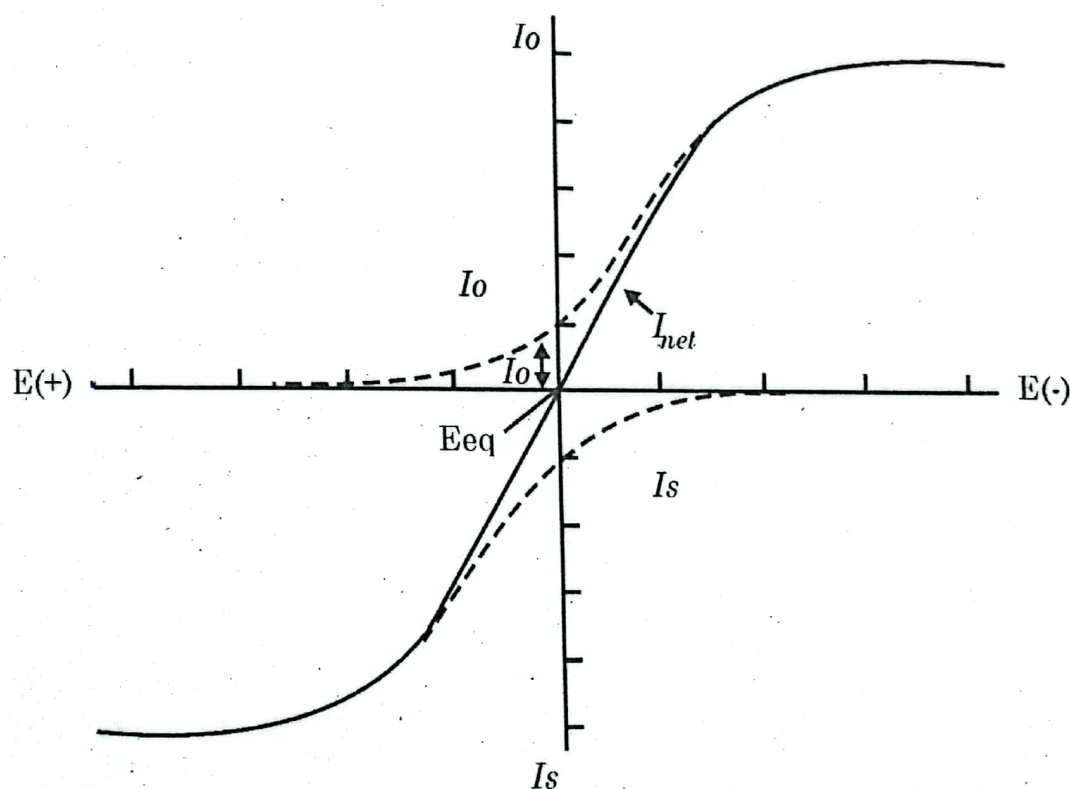


Figure 14.1 Current-potential curve for the system $O + ne \leftrightarrow R$, assuming that electron transfer is rate-limiting, $C_O = C_R$, and $\alpha = 0.5$. The dotted lines show the i_c (cathodic current) and i_a (anodic current) components.

14.3. POLARIZABLE, NON-POLARIZABLE AND SOLID ELECTRODES

Polarizable electrodes are those electrodes in which no actual charge crosses the electrode-electrolyte interface when a current is applied. The current which across the

interface is a displacement current and the electrode behaves like a capacitor ($\text{---}||\text{---}$). A well-known example of polarizable electrode is platinum electrode. Non-polarizable electrodes are those electrodes in which current flows freely across the electrode-electrolyte interface, requiring no energy to make the transition ($\text{---}\text{Z}\text{---}$). The best example of non-polarizable electrode is Ag/AgCl electrode.

The limited anodic potential range of mercury electrodes has precluded their utility for monitoring oxidizable compounds. Accordingly, solid electrodes with extended anodic potential windows have attracted considerable analytical interest. Of the many different solid materials that can be used as working electrodes, the most often used are, carbon, platinum, and gold. Silver, nickel, and copper can also be used for specific applications. Accordingly, the use of such electrodes requires precise electrode pretreatment and polishing to obtain reproducible results. The nature of these pretreatment steps depends on the materials involved. Mechanical polishing (to a smooth finish) and potential cycling are commonly used for metal electrodes, while various chemical, electrochemical, or thermal surface procedures are added for activating carbon-based electrodes. Unlike mercury electrodes, solid electrodes present a heterogeneous surface.

Such electrodes consist of a short cylindrical rod of the electrode material embedded in a tightly fitting tube of an insulating material (Teflon, Kel-F, etc.). Other configurations of solid electrodes, including various ultra-microelectrode and micro-fabricated screen-printed strips or silicon-based thin-film chips, are attracting increasing attention.

✓ 14.4. CYCLIC VOLTAMMETRY

One of the more complex electrochemical techniques, cyclic voltammetry is very frequently used because it offers a wealth of experimental information and insights into both the kinetic and thermodynamic details of many chemical systems. Cyclic voltammetry consists of scanning linearly the potential of a stationary working electrode (in an unstirred solution), using a triangular potential waveform (Figure 14.2). Depending on the information sought, single or multiple cycles can be used. During the potential sweep, the potentiostat measures the current resulting from the applied potential. The resulting current-potential plot is termed a cyclic voltammogram. The cyclic voltammogram is a complicated, time-dependent function of a large number of physical and chemical parameters.

Cyclic voltammetry is often the first experiment performed in an electroanalytical study. In particular, it offers a rapid location of redox potentials of the electroactive species, and convenient evaluation of the effect of media on the redox process. Figure 14.2 illustrates that only the oxidized form O is present initially. Thus, a negative-going potential scan is chosen for the first half-cycle, starting from a value where no reduction occurs.

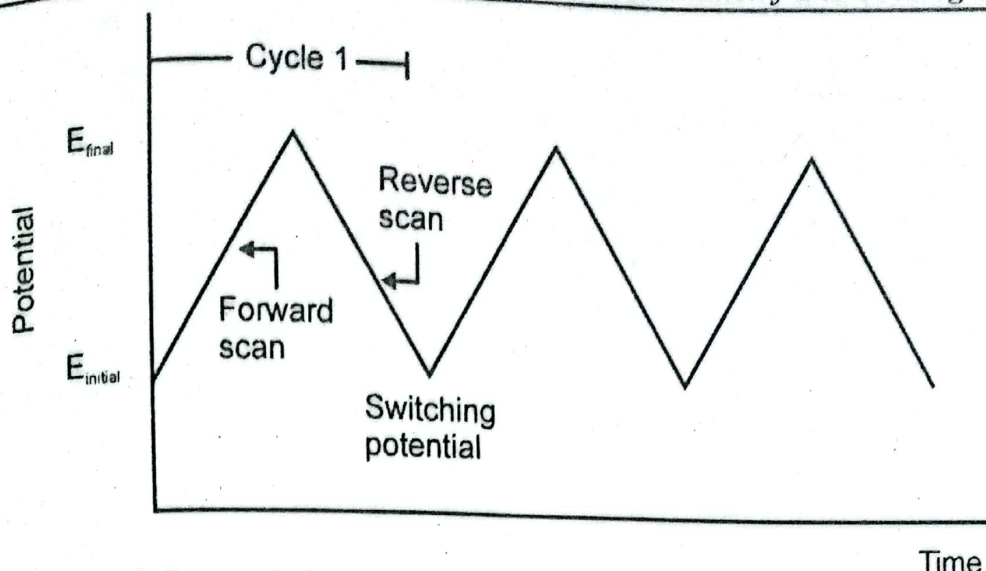


Figure 14.2. Potential-time excitation signal in a cyclic voltammetric experiment.

The characteristic peaks in the cyclic voltammogram are caused by the formation of the diffusion layer near the electrode surface. These can be best understood by carefully examining the concentration-distance profiles during the potential scan. The most common experimental configuration for recording cyclic voltammogram consists of an electrochemical cell that has three electrodes, i.e., counter or auxiliary electrode (C), reference electrode (R), and working electrode (W), all immersed in a liquid and connected to a potentiostat. The potentiostat allows the potential difference between the reference and working electrode to be controlled with minimal interference from IR (ohmic) drop. In this configuration, the current flowing through the reference electrode also can be minimized thereby avoiding polarization of the reference electrode and hence keeping the applied potential distribution between the working and reference electrode stable. Positioning the reference electrode close to the working electrode further helps to minimize the IR drop between the reference and working electrode due to the resistivity of the solution phase.

Reference electrode: Commonly used are aqueous Ag/AgCl or calomel half cells, which can be obtained commercially or easily prepared in the laboratory.

Counter electrode: Non-reactive high surface area electrode, commonly platinum gauze.

Working electrode: Most commonly used are inlaid disc electrodes (Pt, Au, graphite, glassy carbon, etc.) of well-defined area. Other geometries may be employed beneficially in appropriate circumstances (dropping or hanging mercury hemisphere, cylinder, band, arrays, or grid electrodes).

Data Interpretation

The cyclic voltammogram is characterized by several important parameters. Four of these observables, the two peak currents and two peak potentials, provide the basis for the diagnostics developed by Nicholson and Shain for analyzing the cyclic voltammetric response.

14.5. STRIPPING VOLTAMMETRY

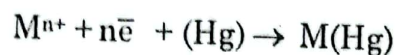
Stripping voltammetry is an extremely sensitive electrochemical technique for measuring trace metals. Its remarkable sensitivity is attributed to the combination of an effective pre-concentration step with advanced measurement procedures that generate an extremely favorable signal background ratio. Since the metals are pre-concentrated into the electrode by factors of 100–1000, detection limits are lowered by two to three orders of

magnitude compared to solution-phase voltammetric measurements. Hence, four to six metals can be measured simultaneously in various matrices at concentration levels down to 10^{-10} M, utilizing relatively inexpensive instrumentation. The ability to obtain such low detection limits strongly depends on the degree to which contamination can be minimized. Expertise in ultra-trace chemistry is required.

Essentially, stripping analysis is a two-step technique. The first, or *deposition step*, involves the electrolytic deposition of a small portion of the metal ions in solution into the mercury electrode to pre-concentrate the metals. This is followed by the *stripping step* (the measurement step), which involves the dissolution (stripping) of the deposit. Different versions of stripping analysis can be employed, depending on the nature of the deposition and measurement steps.

1. Anodic stripping voltammetry (ASV)

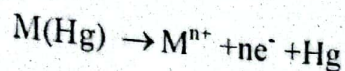
ASV is the most widely used form of stripping analysis. In this case, the metals are being pre-concentrated by electrodeposition into a small-volume mercury electrode (a thin mercury film or hanging mercury drop). The pre-concentration is done by cathodic deposition at a controlled time and potential. The deposition potential is usually 0.3–0.5 V more negative than E° for the least easily reduced metal ion to be determined. The metal ions reach the mercury electrode by diffusion and convection, where they are reduced and concentrated as amalgams:



The convective transport is achieved by electrode rotation or solution stirring (in conjunction with the mercury film electrode) or by solution stirring (when using the hanging mercury drop electrode). Quiescent solutions can be used when using mercury ultra-microelectrodes. The duration of the deposition step is selected according to the concentration level of the metal ions in question, from less than 0.5 min at the 10^{-7} M level to about 20 min at the 10^{-10} M level. The concentration of the metal in the amalgam, C_{Hg} , is given by Faraday's law

$$C_{Hg} = \frac{i_l t_d}{nFV_{Hg}} \quad (14.1)$$

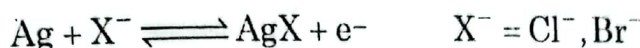
Where, i_l is the limiting current for the deposition of the metal, t_d is the length of the deposition period, and V_{Hg} is the volume of the mercury electrode. The deposition current is related to the flux of the metal ion at the surface. The total amount of metal plated represents a small (and yet reproducible) fraction of the metal present in the bulk solution. Following the preselected time of the deposition, the forced convection is stopped, and the potential is scanned anodically, linearly, or in a more sensitive potential-time (pulse) waveform that discriminates against the charging background current (usually square-wave or differential-pulse ramps). Such pulse excitations also offer reduced oxygen interferences and analyte replating, respectively. During this anodic scan, the amalgamated metals are reoxidized, stripped out of the electrode (in an order that is a function of each metal standard potential), and an oxidation (stripping) current is flowing:



The combination of ultrasound with stripping voltammetry has been proposed to allow the possibility of a wider use of solid electrodes and to overcome electrode passivation in real samples. The major types of interferences in ASV procedures are overlapping stripping peaks caused by a similarity in the oxidation potentials (e.g., of the Pb, Tl, Cd, Sn or Bi, Cu, Sb groups), the presence of surface-active organic compounds that adsorb on the mercury electrode and inhibit the metal deposition, and the formation of intermetallic compounds (e.g., Cu-Zn), which affects the peak size and position.

2. Cathodic Stripping Voltammetry (CSV)

CSV is the mirror image of ASV. It involves anodic deposition of the analyte, followed by stripping in a negative-going potential scan. Cathodic stripping voltammetry is used to measure a wide range of organic and inorganic compounds, capable of forming insoluble salts with mercury. Among these are various thiols or penicillin, as well as halide ions, cyanide, and sulfide. In CSV the deposition and stripping steps involve the following reaction:



The remarkable sensitivity, broad scope, and low cost of *stripping voltammetry* have led to its application in a large number of analytical problems. The technique is useful for the determination of numerous trace metals in environmental, industrial, and clinical samples, as well as for assays of foodstuffs, beverages, gunshot residues, and pharmaceutical formulations.

14.6. INTRODUCTION OF POLAROGRAPHY

The technique polarography may be regarded as a branch of voltammetry in which current voltage relationships are obtained with an electrolysis cell in which one of the electrode is a micro electrode, usually a dropping mercury electrode (DME). The current voltage relationships are then assessed to determine the identity or concentration of species reacting at the micro electrode. Thus it is possible to construct a reproducible current voltage curve if a steady increase in voltage is applied to a cell containing dropping mercury cathode which consists of a mercury reservoir from which mercury trickles down as tiny drops through a capillary into the cell and a pool of mercury at the bottom of the cell as anode.

Polarography may be defined as an instrumental technique which consists in the measurement of potential difference as current flows in solutions and the result so obtained can thus be interpreted in terms of the nature and behavior of many substances and systems. The value of current flowing through the cell is measured with the help of an instrument known as *polarograph* and the curves which are developed from the data are known as *polarograms*. Polarography has several analytical advantages. For example, its wide applicability, quantitative and qualitative determination of various species in the same solution and in the same experiment. It can be used in relatively large concentration range and is quite rapid technique in the sense that quantitative data for a variety of substances can be obtained in from 13-30 minutes. Polarographic analysis using mercury electrodes has a number of advantages. For example, (i) each mercury drop represents a well reproducible cathode with fresh, clean surface, the electrolysis products being removed together with the previous drop falling off the capillary. (ii) The reduction of H^+ ions at the Hg electrode occurs with more negative potentials than at the platinum electrode because of large over voltage for hydrogen formation on mercury.

Basic Principle of Polarography

It has been observed that when a solution containing various ions is electrolyzed with a steady increase in voltage, ions with lower discharge potentials are electrolyzed first. In the presence of those whose discharge potentials are higher. Hence, a concentration gradient will be set up, if the electrolysis is carried out without stirring the electrolyte. This is the basic principle of electroanalytical technique "polarography". Figure 14.3 shows a polarographic curve indicating metal ions having different discharging potentials which are climbing up on the curve accordingly.

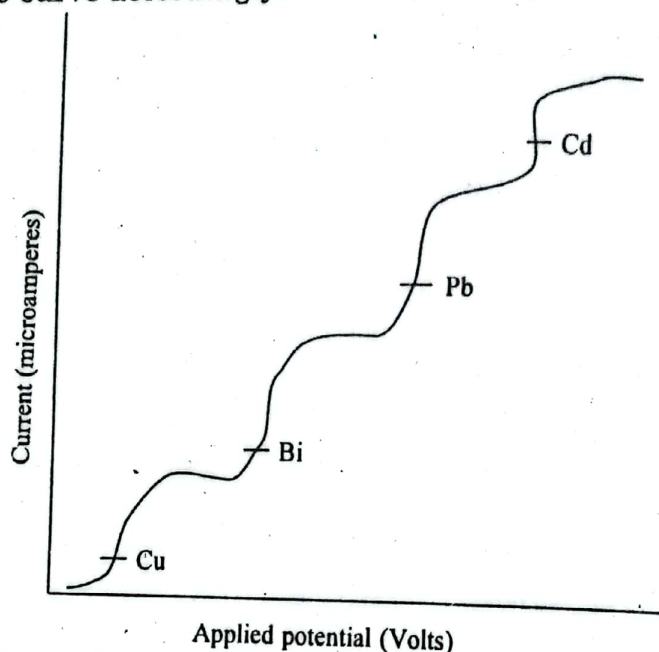


Figure 14.3. Shows a polarographic curve of a solution containing a mixture of metal ions. The curve shows that different metals are arranging themselves according to their discharging potentials which is the basic principle of polarography.

14.7. INTERPRETATION OF VARIOUS POLAROGRAPHIC CURVES

Two polarographic curves are shown in figure 14.4. Curve *b* belongs to polarogram of dilute hydrochloric acid (1 Formal solution) and curve *a* belongs to the mixture of dilute hydrochloric acid (1 Formal solution) and dilute cadmium chloride (0.5 mM solution). No detectable changes of current resulted at potentials less negative than about 0.5 V from the addition of Cd ions. At potential more negative than 0.5 V, a wave appeared indicating the reduction of Cd ions. This process became more and more rapid as Cd ions got diffused from the bulk of the solution upto the surface of the electrode. The rate of this diffusion depends upon a number of factors. However, the concentration of Cd ions in the solution is of great importance. More concentrated is the solution; the greater is the rate at which Cd ions reach the electrode surface resulting a greater diffusion current that results from their reduction. The ease of reduction or oxidation differs for different substances and is reflected by position of the wave with respect to the potential axis. This is termed by a parameter called the *half wave potential* ($-E_{1/2}$).

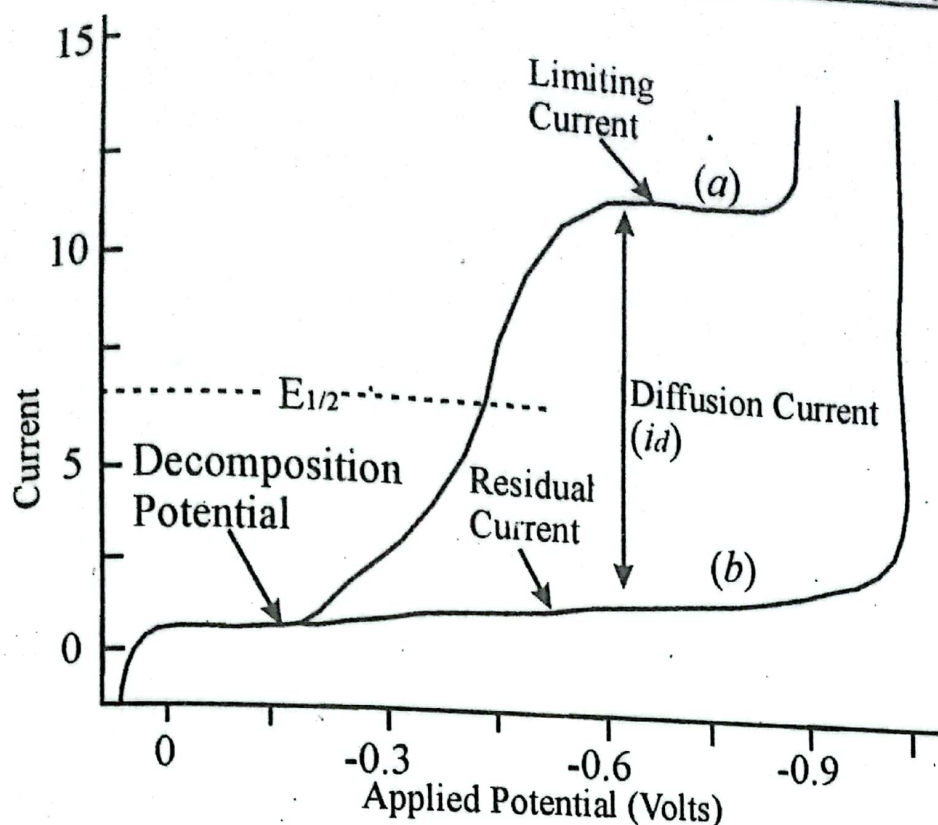


Figure 14.4 Polarograms of (a) 1F HCl with 0.5 mM Cd ions and (b) 1F HCl alone.

This may be defined as the potential at which the current due to the reduction and oxidation of the substance responsible for the wave, is half as large as on the plateau. The half wave potential of Cd ion is shown in Figure 14.4a. Within a set of defined experimental conditions each ion has its own characteristic half wave potential which is the basis of qualitative polarographic analysis.

Decomposition Potential

The voltage at which the polarogram for the electrode reactive species departs from the residual current is known as decomposition potential. In figure 14.4 the decomposition potential has been indicated.

4.8 DIFFUSION, LIMITING AND RESIDUAL CURRENTS

The region in which the current level becomes a smooth line after a sudden rise is known as *limiting current* (Figure 14.2a). The limiting current then becomes totally independent of the voltages applied. In solutions having low electrolytic concentration the portion of *limiting current* that is due to electrostatic forces is known as *migration current*. The migration current is usually eliminated by using a supporting electrolyte whose concentration exceeds that of the reactive species by a factor of 50-100. Under these conditions, the fraction of current carried through the solution by the reactive species is so small to neglect. This is because of the presence of large excess of interfering particles having same charge. The *limiting current* then becomes the *diffusion current* (Figure 14.2). *Residual current* is that current which is due to supporting electrolyte only which passes through the cell even in the absence of Cd ions.

14.9 THE ILKOVIC EQUATION

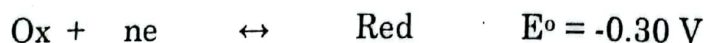
The diffusion current is expressed in the form of Ilkovic equation. It is a relation which is used in polarography relating the diffusion current (i_d) and concentration of depolarizer (c), which is the substance reduced or oxidized at dropping mercury electrode (DME). The Ilkovic equation is represented as

$$i_d = knD^{1/3}m^{2/3}t^{1/6}c \quad (14.2)$$

Where k is a constant which includes Faraday constant, F and density of mercury. The value of k has been evaluated at 708 for max current and 607 for average current. D is the diffusion coefficient of the depolarizer in the medium (cm^2/s), n is the number of electrons exchanged in the electrode reaction, m is the mass flow rate of Hg through the capillary (mg/sec), and t is the drop lifetime in seconds, and c is depolarizer concentration in mol/cm^3 . The equation has been named after the scientist who derived it, the Slovak chemist, Dionýz Ilkovič. If several factors of Ilkovic equation are known the concentration of specie can be measured computationally from the observed current. Only value of i_d cannot be measured easily. However, this problem has been compensated by utilizing the value of conductance of solution.

14.10 LOGARITHMIC ANALYSIS OF POLAROGRAPHIC WAVE

A typical polarographic reduction can be represented as following equation



Where, Ox represents the substance which undergoes reduction at working electrode and Red represents the product of reduction. The reduced material may be a metal amalgam, a soluble ion, a metallic deposit on the surface of a micro-electrode or a molecule. Let us consider -0.30 V a standard potential. Now let us consider possible chemical changes occurring in the film of liquid immediately surrounding the surface of microelectrode. The potential of the electrode thus be calculated by Nernst equation.

$$E_{\text{cathode}} = E^\circ - [0.0591/n] \log[\text{Red}]_o/[\text{Ox}]_o \quad (14.3)$$

The applied potential may be expressed as

$$E_{\text{applied}} = E_{\text{cathode}} - E_{\text{anode}} \quad (14.4)$$

Where $E_{\text{anode}} = E_{\text{SCE}}$, processing the above equation

$$E_{\text{applied}} = E^\circ - [0.0591/n] \log[\text{Red}]_o/[\text{Ox}]_o - E_{\text{SCE}} \quad (14.5)$$

Equation (14.5) may be used to calculate the ratio between the activities of reactants and products at the surface of electrodes for any applied potential.

For example, if, $n = 1$, $E_{\text{SCE}} = 0.2422$ and $E_{\text{applied}} = -0.06 \text{ V}$

We get following ratio from equation (14.5)

$$[\text{Ox}]_o/[\text{Red}]_o = 0.1$$

This ratio continues to decrease at higher potentials. For example, at -0.9 V the ratio value is 1×10^{-6} and at -1.0 V it becomes 2×10^{-8} . Hence, as the concentration of Ox at surface of the electrode approaches zero, the expression for current becomes,

$$i_d = k[\text{Ox}] \quad \text{when } [\text{Ox}]_o \ll [\text{Ox}] \quad (14.6)$$

At large potentials, rate of diffusions thus constant and hence current is also constant. The magnitude of the diffusion current is directly proportional to the concentration of reactant in bulk of the solution. This fact is the foundation of quantitative polarography. When a limiting current is achieved as a consequence of the limitation of rate at which a reactant can be brought to the surface of electrode, a state of complete concentration polarization is said to exist.

14.11. ADVANTAGES AND LIMITATIONS OF DROPPING MERCURY ELECTRODE (DME)

DME possesses several unique and useful advantages.

1. The surface area of DME is reproducible with any given capillary. During the growth of each drop the increase in area of the drop and the outward movement of its surface more than compensates for the tendency observed with stationary electrode, to exhibit a decreasing current with time. At a constant applied potential, a steady average current, which is easily measured (and its proportional to the concentration of reacting species) may be observed for successive drops.
2. The constant renewal of electrode surface eliminates passivity or poisoning effect.
3. There is large hydrogen over voltage on Hg. Therefore, reduction of even the alkali metal ions can be observed.
4. Mercury forms solid solution that is amalgams with many metals and hence, makes many metal ions more easily reducible.
5. The surface area can be calculated from the weight of the drop.
6. The diffusion current assumes steady value immediately and thus is reproducible.
7. A new metal surface is continuously generated; the behavior of the electrode is thus independent of its past history.

There are only few limitations of using DME. These are as following.

1. It oxidizes easily. This property severely restricts the use of mercury as an anode. Thus DME can be used only for analysis of reproducible or very easily oxidisable substances.
2. It is much more limited to its voltages range in the positive direction than are the Nobel metals, such as platinum and gold. The potential at which the anodic dissolution of Hg starts depends upon the identity and concentration other specie present in the solution.